

PUKASKWA, ON, Canada

WMO: 717500

Lat: 48.608N

Lon: 86.287W

Elev: 208

StdP: 98.86

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-26.7	-33.6	0.2	-29.9	-30.3	0.2	-26.3	16.1	-4.3	14.2	-3.3	2.8	50	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	23.1	17.5	21.6	17.4	20.4	16.8	19.4	21.4	18.5	20.4	17.7	19.5	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.8	20.3	17.8	13.1	19.5	16.9	12.4	18.7	56.2	21.3	53.4	20.5	50.8	19.5	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.7	11.0	9.3	-33.6	27.3	4.2	1.7	-36.6	28.5	-39.1	29.5	-41.5	30.5	-44.6	31.7		
			-33.7	21.2	4.2	1.2	-36.7	22.0	-39.2	22.8	-41.6	23.4	-44.6	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-12.7	-10.5	-5.0	1.5	7.6	12.0	14.4	15.6	12.3	5.9	-1.3	-8.4
	DBStd	11.04	7.84	7.09	6.53	4.47	3.94	3.23	3.07	2.81	3.94	4.27	5.81	7.29
	HDD10.0	3186	704	574	466	255	97	16	3	1	22	142	338	569
	HDD18.3	5724	963	807	723	504	333	192	126	91	184	386	588	828
	CDD10.0	518	0	0	1	1	22	74	140	174	91	14	1	0
	CDD18.3	15	0	0	0	0	0	1	4	7	2	0	0	0
	CDH23.3	46	0	0	0	0	2	11	17	13	3	0	0	0
	CDH26.7	5	0	0	0	0	0	2	1	2	1	0	0	0
Wind	WSAvg	3.8	4.2	4.1	3.9	3.6	3.3	2.9	2.9	3.5	4.1	4.3	4.4	4.3
Precipitation	PrecAvg	891	59	45	48	66	83	90	87	86	105	94	67	61
	PrecMax	1067	126	88	83	135	121	162	151	140	172	140	118	101
	PrecMin	688	38	14	18	31	38	37	23	35	39	44	38	30
	PrecStd	96	18	17	16	24	20	27	31	27	32	30	19	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	5.3	13.7	15.9	22.3	24.7	25.6	24.8	23.0	18.2	12.4	5.8
		MCWB	3.3	3.4	9.7	9.5	14.0	16.5	18.0	18.0	18.5	14.7	10.1	4.8
	2%	DB	2.3	3.1	7.7	12.5	18.4	21.5	22.8	22.6	20.9	16.1	10.0	4.5
		MCWB	1.7	2.1	5.6	7.2	13.5	15.3	17.7	18.3	17.6	13.8	8.6	3.6
	5%	DB	1.2	1.5	5.2	10.1	16.1	19.4	21.1	21.4	19.6	14.6	8.1	3.5
		MCWB	0.6	0.7	3.3	5.6	11.8	14.8	17.1	17.8	17.1	12.7	6.8	2.6
	10%	DB	-0.4	0.1	3.2	8.1	13.9	17.6	19.6	20.4	18.3	12.7	6.7	1.9
		MCWB	-1.0	-0.8	1.6	4.5	9.9	13.8	16.5	17.3	16.3	11.1	5.6	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.9	10.2	10.6	17.2	19.1	20.4	20.8	19.6	16.1	10.9	5.2
		MCDB	3.5	4.7	12.3	13.2	18.4	21.8	22.7	22.6	21.5	17.2	11.5	5.5
	2%	WB	1.8	2.1	5.9	8.2	14.6	17.3	18.9	19.6	18.6	14.1	8.6	3.8
		MCDB	2.2	2.8	7.4	11.4	17.3	19.7	21.2	21.5	20.1	15.6	9.7	4.4
	5%	WB	0.6	0.8	3.6	6.4	12.4	15.9	18.0	18.7	17.6	12.7	7.1	2.6
		MCDB	1.1	1.4	4.9	9.3	15.2	18.0	20.1	20.6	19.1	14.3	8.0	3.4
	10%	WB	-1.1	-0.7	1.9	4.8	10.4	14.5	17.0	17.9	16.6	11.3	5.8	1.2
		MCDB	-0.4	0.1	3.0	7.4	13.0	16.7	19.0	19.6	18.1	12.7	6.7	1.9
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.5	10.3	9.0	9.1	9.3	8.1	7.8	7.7	6.8	6.6	8.5
		MCDBR	7.2	7.6	9.0	11.6	12.2	12.0	10.3	8.7	7.9	8.3	6.9	7.2
		MCWBR	7.1	6.9	6.6	7.3	7.5	7.0	5.9	5.2	5.4	6.1	6.2	7.1
	5% WB	MCDBR	7.3	7.5	8.2	10.1	10.9	10.3	8.5	7.2	7.0	7.4	6.8	7.2
		MCWBR	7.4	7.1	6.3	7.1	7.9	7.6	5.9	4.9	5.4	5.9	6.2	7.2
Clear-Sky Solar Irradiance	taub	0.269	0.270	0.306	0.341	0.363	0.370	0.387	0.369	0.360	0.325	0.289	0.262	
	taud	2.396	2.407	2.366	2.357	2.402	2.404	2.394	2.454	2.463	2.529	2.528	2.454	
	Ebn at Noon	810	900	912	909	898	891	869	872	844	823	787	777	
	Edh at Noon	65	81	100	113	113	113	113	102	92	73	58	54	
All-Sky Solar Radiation	RadAvg	1.08	2.09	3.50	4.86	5.45	5.89	5.87	5.01	3.57	2.09	1.04	0.78	
	RadStd	0.14	0.20	0.31	0.45	0.39	0.41	0.43	0.40	0.25	0.25	0.11	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page